DEPARTMENT OF
SOFTWARE TECHNOLOGY

STALGCM

Advanced Algorithms and Complexities

MCO 2: Project

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Course Site/Repository : AnimoSpace, Google Drive

Description

Aligned with the course objectives, students are expected to develop formal machines that address existing research and societal problems. In line with this, one of the Major Course Outputs (MCOs) is a project focusing on spell checking – an application aligned with United Nations Sustainable Development Goal 4 (Quality Education). Through this task, students will analyze the problem domain, design and implement computational models that demonstrate the applicability of abstract models and formal languages to real-world contexts.

Real-world context

We recognize that spelling rules exist (e.g., *o* → *u* substitutions), but such rules alone are insufficient to accurately detect spelling errors.
For example:

- kumpleto → kompleto → spelling error
- upo → opo → *not* a spelling error

Your task is to design three formal rules that, when applied to a corpus, produce a set of detected spelling errors with 100% precision. That is, all words identified by your rules as spelling errors must indeed be incorrect (i.e., no false positives).

Below are some examples (n.b., These examples may not be used for the project.):

- umC → omC (kumpleto → kompleto)
- CwV → CuwV (kwento → kuwento)
- CyV → CiyV (pyano → piyano)

Where:

- V = vowels [aeiou]
- C = consonants

Points

This MCO is worth 100 points.
Bonus: Integrated in MS Word: +10 points
Perfect score: 110 / 100

Tasks

Group Size: Students will work in groups of up to three (3) members. Cross-section groupings are allowed.

Tasks:

1. Research. Read authoritative sources on Filipino spelling guidelines – Ortograpiyang Pambansa (OP) and Manwal sa Masinop na Pagsulat (MMP). Your rules must be grounded in documented orthographic principles. For the purposes of this project, one form must be demonstrably incorrect according to OP or MMP in order to be classified as a spelling error.
2. Rule Development. Develop three formal spelling rules and integrate them into LanguageTool, a rule-based grammar and spell-checking engine. Ensure that:
    a. Rules are precisely defined.
    b. Exceptions are properly handled.
    c. The system does not produce false positives (100% precision requirement).
3. Documentation. For each spelling error, provide three (3) examples and identify two (2) websites or platforms where the incorrect spelling appears. This results in a minimum of six (6) documented occurrences per rule.
4. If Generative AI (GenAI) tools are used at any stage (e.g., brainstorming, drafting, coding assistance), their use must be clearly disclosed in the documentation.

Example: CwV → CuwV

- Three examples: Kwento → Kuwento; Linggwahe → Lingguwahe; Kwadra → Kuwadra
- Site 1 (kwento): https://web.facebook.com/MalilikotnaKwento/
- Site 2 (kwento): https://www.youtube.com/c/mgakwentonithelmo

Declaration of AI usage (per person):

1. AI tool(s) used
2. Describe briefly how you used the AI tool(s) in the completion of this assessment task. Examples include: idea generation, grammar checking, paraphrasing, summarizing references, formatting help, etc.
3. Extent of Use:
    a. Minimal – used only for grammar/spell check.
    b. Moderate – used for inspiration, then reworded significantly.
    c. Extensive – relied heavily on AI-generated content with minimal modification.
4. In 2–3 sentences, reflect on how the use of AI contributed to your learning. If no AI was used, reflect on why you chose not to.

Deadline

March 31, 11:59 pm.
Deliverables:

1. Rule File (.xml)
    a. Submit a newly developed rule file created from scratch.
    b. The file must be functional and compatible with LanguageTool.
2. Documentation (.pdf)
    a. Include the names, section, and group number of all members.
    b. Provide documentation for the three rules, including three (3) examples per rule and two (2) websites or platforms per rule where the incorrect spelling appears.
    c. Indicate the LanguageTool version used and include a screenshot of the MS Word plugin functioning (if applicable).
    d. Table of contribution.
    e. Include a declaration of AI usage (per person).
3. Video Demonstration
    a. Provide a short demo showing the rules running in the standalone version and, if applicable, as a plugin for MS Word.
    b. Discuss each rule and demonstrate example outputs.
    c. Explain how exceptions are handled.
    d. All members must appear on camera in the video and actively participate in the presentation. The presentation should be spontaneous (i.e., not fully scripted or read verbatim).

Rubric

Criteria	**Exemplary** Full points	**Satisfactory** ¾ points	**Developing** ½ points	**Beginning** ¼ points
Rule development (25 points)	Rules for three spelling errors were developed.	Rules for two spelling errors were developed.	Rule for one spelling errors was developed.	Rules were defined but not integrated in LanguageTool
Precision (25 points)	Exceptions are properly handled ensuring precision.	Minor false positives observed.	The system does not mark most correct spelling as incorrect.	The system marks some incorrect spelling as incorrect.
Technical strength (25 points)	The rule file also incorporates exceptions attribute.	The rule file incorporates regular expressions both in the pattern and in the suggestion.	The rule file also incorporates regular expressions in the pattern.	The rule file includes a short and at least two examples (one correct and one incorrect).
Documentation and demo (25 points)	Demo video is spontaneous, and the student(s) appear confident.	Documentation includes websites or platforms that contain incorrect spelling.	Demo video presentation appears to be read.	Documentation is 50% complete.

Bonus: Integrated in MS Word: +10 points

Resources

- Exploration activity: Canvas → Assignment → Exploration: LanguageTool
- LanguageTool GDrive: GDrive → Resources → LanguageTool
- Reference materials: GDrive → Filipino language → MMP and OP
- LanguageTool dev: https://dev.languagetool.org/
- Applied in spell checking: https://aclanthology.org/2024.paclic-1.141.pdf
- Applied in grammar checking: https://aclanthology.org/W11-3402.pdf
- Applied in code-switching point detection: https://aclanthology.org/Y12-1024.pdf